

Lab 08: Recurrent Neural Networks (RNN & LSTM)

Objective

In this lab, you will explore different Recurrent Neural Network architectures and compare their performance on a sentiment analysis task using the IMDb dataset.

Task 1: Build the Baseline Model

Run the provided **SimpleRNN** model.

Record the following:

- Training Accuracy
- Validation Accuracy
- Test Accuracy
- Training Time

Questions

1. What is the role of the Embedding layer?
 2. Why do we use SimpleRNN instead of a Dense layer?
 3. What is the purpose of the Sigmoid activation function?
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Task 2: Increase the Number of RNN Units

Modify the model by changing the number of units from **32** to **64**.

Compare the new model with the baseline.

Record:

- Test Accuracy
- Training Time

Questions

1. Did the model performance improve?
 2. Did the training time increase?
 3. Explain your observations.
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Task 3: Build a Stacked RNN

Create a model with **two SimpleRNN layers**.

Hint: Use `return_sequences=True` in the first RNN layer.

Record:

- Test Accuracy
- Training Time

Questions

1. Why is `return_sequences=True` required?
 2. Did stacking RNN layers improve the results?
 3. Is adding more layers always beneficial? Explain.
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Task 4: Build a Bidirectional RNN

Replace the SimpleRNN layer with a **Bidirectional SimpleRNN**.

Record:

- Test Accuracy
- Training Time

Questions

1. Compare the results with the baseline model.
 2. Why might Bidirectional RNN achieve better performance?
 3. Is Bidirectional RNN suitable for real-time next-word prediction? Why or why not?
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Task 5: Replace SimpleRNN with LSTM

Build an LSTM model using the same architecture.

Record:

- Test Accuracy
- Training Time

Questions

1. Compare the performance with SimpleRNN.
 2. Which model required more training time?
 3. Why is LSTM generally better for long text sequences?
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Task 6: Build a Bidirectional LSTM

Create a Bidirectional LSTM model.

Record:

- Test Accuracy
- Training Time

Questions

1. Which model achieved the highest accuracy?
 2. Which model required the longest training time?
 3. Was the performance improvement worth the additional complexity? Explain.
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Task 7: Hyperparameter Experiment

Modify **ONE** of the following hyperparameters:

- Embedding dimension
- Number of RNN/LSTM units
- Maximum sequence length (maxlen)
- Batch size

Record the effect on:

- Test Accuracy
- Training Time

Explain your observations.

Results Table

Model	Test Accuracy	Training Time	Observation
SimpleRNN			
SimpleRNN (64 Units)			
Stacked RNN			
Bidirectional RNN			
LSTM			
Bidirectional LSTM			

Bonus Challenge

Design your own sentiment analysis model using any combination of the following layers:

- Embedding
- SimpleRNN
- LSTM
- Bidirectional
- Multiple RNN/LSTM layers

Your goal is to achieve the highest possible **Validation Accuracy** while keeping the model as simple as possible.

In your report, explain:

- Why you selected this architecture.
- What modifications you made.
- Whether your model outperformed the baseline model.